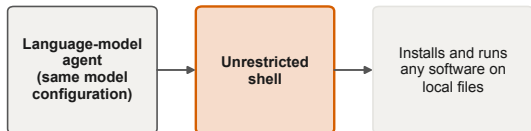


a

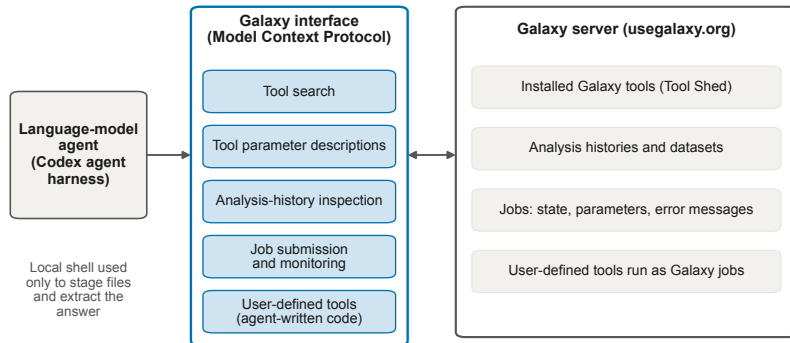
### Open-ended code condition (reference)



Execution trace: commands, scripts, command output and final files; no record of individual analysis jobs

b

### Galaxy condition (Galaxy-mediated execution)



Execution trace and analysis history: tool identities and versions, requested and resolved parameters, datasets, job states and error messages

c

### Three benchmarks, five model configurations, three replicate runs of every task

| Benchmark                                                      | BixBench-Verified-50 |        | CompBioBench    |        | IWC             |        |
|----------------------------------------------------------------|----------------------|--------|-----------------|--------|-----------------|--------|
|                                                                | Open-ended code      | Galaxy | Open-ended code | Galaxy | Open-ended code | Galaxy |
| Execution condition                                            |                      |        |                 |        |                 |        |
| GPT-5.5                                                        | 150                  | 150    | 300             | 300    | 30              | 30     |
| GPT-5.6 Sol                                                    | 150                  | 150    | 300             | 300    | 30              | 30     |
| GPT-5.6 Luna                                                   | 150                  | 150    | 300             | 300    | 30              | 30     |
| DeepSeek V4 Pro (Codex)                                        | 150                  | 150    | 300             | 300    | 30              | 30     |
| DeepSeek V4 Pro (Claude Code, superseded; reported separately) | 150                  | 150    | —               | —      | —               | —      |
| GPT-6 Astra (open-ended code condition only; not paired)       | —                    | —      | 100             | —      | —               | —      |

Numbers are runs: tasks × 3 replicate runs, per model configuration and execution condition

#### BixBench-Verified-50

50 tasks; platform-neutral benchmark  
Endpoint: accuracy (share of runs scored correct by the benchmark evaluator)

#### CompBioBench

100 tasks; platform-neutral benchmark  
Endpoint: reported benchmark score (answers credited, of 100)

#### IWC (Intergalactic Workflow Commission)

10 tasks; workflow-derived benchmark  
Endpoint: output agreement with the workflow's reference output (0 to 1)

Archive: 4,240 runs · 4,228 execution traces  
23,080 Galaxy analysis jobs · 69,812 Galaxy interface calls